Review of Submission 3B

First Proof Second Batch
Reviewer 1
June 4–8, 2026

1 Review

Correctness. The model arrives at the conjecturally correct range of probabilities, but falsely claims a complete proof. The counterexamples for $p \in (1/3, 1/2)$ and $p \in (1/2, 1)$ are given correctly. The cases of $p - 0, 1$ and the short argument via symmetry at $p = 1/2$ are also presented. However, the proof for the remaining interval $p \leq 1/3$ is reduced to two inputs, both of which are falsely concluded by referencing the literature. The reductions are correct, but to the best of my knowledge both inputs are plausible but proofs remain open.

Novelty. The novelty of the argument is limited at best. The reduction to the two inputs is rather naive, and the conclusion is incorrectly made without much further work. The correct cases are by its own admission elementary. There are also detailed arguments to deduce the inupts from the falsely cited literature, but they appear to be routine. (In hindsight, the main connection to the Manickam-Miklos-Singhi conjecture is not made.)

Presentation. The presentation of the paper, aside from the incorrect references, is not bad. An overview of the proof strategy and main ideas is given at the beginning, and there is some context shared around each of the citations. The remainder of the argument is a series of routine computations where explanations would improve flow but are not necessary for comprehension.

2 Recommendation

I recommend a decision of rejection. While the argument does reduce the problem to two concrete inputs, it is not clear that the inputs are any easier than the original question. This combined with the lack of novelty in the reduction leads to my recommendation.

3 Submission with inline comments

Overview / Strategy

The problem is to determine for which probabilities p the displayed inequality holds uniformly over all finite convex weight vectors. The main positive argument has two reductions. First, Hoeffding's extremal theorem for convex combinations of i.i.d. random variables reduces the weighted problem to the special case of equal weights. In the Bernoulli case, equal weights give binomial averages. Second, a classical binomial mean-tail estimate of Jogdeo–Samuels type says that for $0 \le p \le 1/3$, every binomial average is at least its mean with probability at least p. These two facts are stated precisely in Section 1 and used in Proposition 5.

The remaining cases are elementary. At $p = 1/2$, the Bernoulli subset is uniformly distributed over all subsets, and pairing every subset with its complement gives probability at least $1/2$. At $p = 1$, the assertion is immediate. To show that there are no other probabilities, equal weights on three variables give counterexamples for every $p \in (1/3, 1/2)$, and equal weights on two variables give counterexamples for every $p \in (1/2, 1)$.

Definition 1. A Bernoulli(p) random variable is a random variable taking value 1 with probability p and value 0 with probability $1 - p$. For $p \in [0, 1]$, let $\mathcal{P}(p)$ denote the following assertion: for every integer $m \ge 1$, every choice of nonnegative real numbers

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and every family $v_1, \dots, v_m$ of i.i.d. Bernoulli(p) random variables, one has

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p.$$

Theorem 2 (Headline theorem: the original problem)**.** *The assertion $\mathcal{P}(p)$ holds exactly for*

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

Equivalently, the complete set of probabilities p for which

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p$$

holds for every finite convex weight vector $(w_1, \dots, w_m)$ and every i.i.d. Bernoulli(p) family $v_1, \dots, v_m$ is

$$\boxed{[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}}.$$

3.1 Classical one-dimensional inputs

The proof uses two standard one-dimensional results. They are quoted in the precise forms needed below. The first is Hoeffding's reduction theorem for convex combinations of i.i.d. random variables; the second is the binomial mean-tail estimate used to evaluate the equal-weight case.

Lemma 3 (Hoeffding's reduction for convex combinations)**.** *Let $X_1, X_2, \ldots$ be i.i.d. real-valued random variables. For $t \in \mathbb{R}$, define*

$$L(t) := \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq t\right].$$

Then for every integer $m \geq 1$ and every choice of weights

$$w_1, \ldots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

one has

$$\Pr\left[\sum_{i=1}^{m} w_i X_i \geq t\right] \geq L(t).$$

Proof. This is the one-dimensional half-line form of Hoeffding's extremal theorem for convex combinations of i.i.d. random variables; see Hoeffding [1]. To make explicit that the closed-tail convention used here causes no ambiguity, we record the standard passage from the open-tail formulation.

Hoeffding's theorem gives, for every real u,

$$\Pr\left[\sum_{i=1}^{m} w_i X_i > u\right] \geq \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} > u\right].$$

Fix $t \in \mathbb{R}$. For every $\varepsilon > 0$, apply the preceding inequality with $u = t - \varepsilon$. For each N,

$$\left\{\frac{X_1 + \cdots + X_N}{N} \geq t\right\} \subseteq \left\{\frac{X_1 + \cdots + X_N}{N} > t - \varepsilon\right\},$$

and therefore

$$\inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} > t - \varepsilon\right] \geq \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq t\right] = L(t).$$

Thus

$$\Pr\left[\sum_{i=1}^{m} w_i X_i > t - \varepsilon\right] \geq L(t) \qquad \text{for every } \varepsilon > 0.$$

As $\varepsilon \downarrow 0$, the events

$$\left\{\sum_{i=1}^{m} w_i X_i > t - \varepsilon\right\}$$

decrease to

$$\left\{\sum_{i=1}^{m} w_i X_i \geq t\right\}.$$

By continuity of probability from above,

$$\Pr\left[\sum_{i=1}^{m} w_i X_i \geq t\right] \geq L(t).$$

This is the desired closed-tail statement. □

> **Reviewer 1**
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> The citation [1] does not contain the theorem being used. In fact, the Lemma as stated is false. Consider for example X_i which takes values 0 with probability 0.1, 1 with probability 0.1, and 2 with probability 0.8. The claimed inequality is false for example $t = 0.8$. There may be hope to salvage this for binary random variables X_i.

Lemma 4 (Binomial mean-tail inequality)**.** *Let*

$$X_{N,p} \sim \mathrm{Bin}(N, p).$$

Then the following hold.

1. *If* $0 \leq p \leq \frac{1}{3}$, *then for every* $N \geq 1$,

$$\Pr[X_{N,p} \geq Np] \geq p.$$

   Equivalently, since $X_{N,p}$ *is integer-valued,*

$$\Pr[X_{N,p} \geq \lceil Np \rceil] \geq p.$$

2. *If* $p = \frac{1}{2}$, *then for every* $N \geq 1$,

$$\Pr[X_{N,1/2} \geq N/2] \geq \frac{1}{2}.$$

Moreover, in the first assertion equality occurs for $N = 1$ *and* $0 < p \leq 1/3$.

Proof. For $p = 0$, $X_{N,0} = 0$ almost surely, so

$$\Pr[X_{N,0} \geq 0] = 1 \geq 0.$$

For $0 < p \leq 1/3$, the first assertion is the classical binomial mean-tail estimate of Samuels and Jogdeo–Samuels; see [3, 2]. In the form needed here, that estimate states exactly that for every $N \geq 1$,

$$\Pr[X_{N,p} \geq \lceil Np \rceil] \geq p.$$

Because $X_{N,p}$ is integer-valued, the events

$$\{X_{N,p} \geq Np\} \quad \text{and} \quad \{X_{N,p} \geq \lceil Np \rceil\}$$

are identical. This proves the first assertion.

For $p = 1/2$, the binomial distribution is symmetric:

$$\Pr[X_{N,1/2} = j] = \Pr[X_{N,1/2} = N - j] \qquad (0 \leq j \leq N).$$

If $N = 2r + 1$ is odd, then the values $0, \dots, r$ pair with the values $2r + 1, \dots, r + 1$, and exactly one member of each pair is at least $N/2 = r + \frac{1}{2}$. Hence

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}.$$

If $N = 2r$ is even, then the values $0, \dots, r - 1$ pair with $2r, \dots, r + 1$, while r is the central value and is included in the event $\{X_{N,1/2} \geq r\}$. Therefore

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}\Big(1 - \Pr[X_{N,1/2} = r]\Big) + \Pr[X_{N,1/2} = r] = \frac{1}{2} + \frac{1}{2}\Pr[X_{N,1/2} = r] \geq \frac{1}{2}.$$

This proves the second assertion.

Finally, if $N = 1$ and $0 < p \leq 1/3$, then

$$\{X_{1,p} \geq p\} = \{X_{1,p} = 1\},$$

so

$$\Pr[X_{1,p} \geq p] = p.$$

Thus equality is possible in the first assertion. □

> **Reviewer 1**
>
> The cited papers make relevant conclusions but the lemma is not immediate. For example, [2] is restricted to the case where np is an integer. Either more proof or a better citation is needed.

3.2 The positive cases

Proposition 5 (The interval $0 \leq p \leq 1/3$)**.** *The assertion $\mathcal{P}(p)$ holds for every*

$$p \in [0, \tfrac{1}{3}].$$

Proof. Fix $p \in [0, 1/3]$. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables.

Extend $v_1, \dots, v_m$, if necessary, to an infinite i.i.d. Bernoulli(p) sequence $X_1, X_2, \dots$ with $X_i = v_i$ for $1 \le i \le m$. Applying Lemma 3 with $t = p$ gives

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] = \Pr\left[\sum_{i=1}^{m} w_i X_i \ge p\right] \ge \inf_{N \ge 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \ge p\right].$$

For each N, the sum $X_1 + \cdots + X_N$ has binomial distribution $\mathrm{Bin}(N, p)$. Hence, writing $X_{N,p} \sim \mathrm{Bin}(N, p)$,

$$\Pr\left[\frac{X_1 + \cdots + X_N}{N} \ge p\right] = \Pr[X_{N,p} \ge Np].$$

Therefore

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge \inf_{N \ge 1} \Pr[X_{N,p} \ge Np].$$

By Lemma 4, since $0 \le p \le 1/3$,

$$\Pr[X_{N,p} \ge Np] \ge p \qquad \text{for every } N \ge 1.$$

Taking the infimum over N gives

$$\inf_{N \ge 1} \Pr[X_{N,p} \ge Np] \ge p.$$

Consequently,

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p.$$

Since m, the weights, and the Bernoulli(p) family were arbitrary, $\mathcal{P}(p)$ holds. □

Proposition 6 (The point $p = 1/2$)**.** *The assertion $\mathcal{P}(1/2)$ holds.*

Proof. Let $m \ge 1$, let

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli$(1/2)$ random variables.

Let

$$A = \{i \in \{1, \dots, m\} : v_i = 1\}.$$

Because the v_i are independent and each is 1 with probability $1/2$, the random set A is uniformly distributed over all 2^m subsets of $\{1, \dots, m\}$. For a subset $B \subseteq \{1, \dots, m\}$, write

$$w(B) = \sum_{i \in B} w_i.$$

Then

$$\sum_{i=1}^{m} w_i v_i = w(A),$$

so the desired event is

$$\{w(A) \geq 1/2\}.$$

Pair every subset $B \subseteq \{1, \ldots, m\}$ with its complement B^c. These pairs partition the 2^m subsets into 2^{m-1} unordered pairs. Since

$$w(B) + w(B^c) = \sum_{i=1}^{m} w_i = 1,$$

at least one of B and B^c has weight at least $1/2$. Thus at least one subset in each complementary pair belongs to the event $\{w(A) \geq 1/2\}$.

Every subset has probability 2^{-m}. Since at least 2^{m-1} subsets satisfy $w(B) \geq 1/2$, we obtain

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq \frac{1}{2}\right] = \Pr[w(A) \geq 1/2] \geq 2^{m-1} \cdot 2^{-m} = \frac{1}{2}.$$

Therefore $\mathcal{P}(1/2)$ holds. $\square$

Proposition 7 (The point $p = 1$)**.** *The assertion $\mathcal{P}(1)$ holds.*

Proof. Let $m \geq 1$, let

$$w_1, \ldots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \ldots, v_m$ be i.i.d. Bernoulli(1) random variables. A Bernoulli(1) random variable equals 1 almost surely, so

$$v_i = 1 \qquad \text{almost surely for every } i.$$

Therefore

$$\sum_{i=1}^{m} w_i v_i = \sum_{i=1}^{m} w_i = 1$$

almost surely. Hence

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq 1\right] = 1 \geq 1.$$

Thus $\mathcal{P}(1)$ holds. $\square$

3.3 Counterexamples outside the claimed set

Proposition 8 (Failure for $1/3 < p < 1/2$)**.** *For every*

$$p \in \left(\frac{1}{3}, \frac{1}{2}\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/3, 1/2)$. Take

$$m = 3, \qquad w_1 = w_2 = w_3 = \frac{1}{3},$$

and let v_1, v_2, v_3 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2 + v_3.$$

Then $N \sim \mathrm{Bin}(3, p)$ and

$$\sum_{i=1}^{3} w_i v_i = \frac{N}{3}.$$

Because $p > 1/3$, one success is not enough:

$$\frac{1}{3} < p.$$

Because $p < 1/2 < 2/3$, two successes are enough:

$$\frac{2}{3} \geq p.$$

Thus

$$\left\{\sum_{i=1}^{3} w_i v_i \geq p\right\} = \{N \geq 2\}.$$

Therefore

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] = \Pr[N \geq 2].$$

Since $N \sim \mathrm{Bin}(3, p)$,

$$\Pr[N \geq 2] = \binom{3}{2} p^2(1-p) + p^3 = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p - p^2(3-2p) = p(1 - 3p + 2p^2) = p(1-p)(1-2p).$$

For $p \in (1/3, 1/2)$, the factors p, $1-p$, and $1-2p$ are all positive. Hence

$$p - p^2(3-2p) > 0,$$

or equivalently,

$$p^2(3-2p) < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

Proposition 9 (Failure for $1/2 < p < 1$)**.** *For every*

$$p \in \left(\frac{1}{2}, 1\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/2, 1)$. Take

$$m = 2, \qquad w_1 = w_2 = \frac{1}{2},$$

and let v_1, v_2 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2.$$

Then

$$\sum_{i=1}^{2} w_i v_i = \frac{N}{2}.$$

Because $p > 1/2$, one success is not enough:

$$\frac{1}{2} < p.$$

Since N can only take the values $0, 1, 2$, the event

$$\left\{\frac{N}{2} \geq p\right\}$$

occurs only when $N = 2$, that is, when both v_1 and v_2 equal 1. Hence

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] = \Pr[v_1 = 1,\ v_2 = 1].$$

By independence,

$$\Pr[v_1 = 1,\ v_2 = 1] = p^2.$$

Since $0 < p < 1$,

$$p^2 < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

3.4 Proof of the classification

Proof of Theorem 2. Proposition 5 proves that $\mathcal{P}(p)$ holds for every

$$p \in [0, \tfrac{1}{3}].$$

Proposition 6 proves that $\mathcal{P}(1/2)$ holds, and Proposition 7 proves that $\mathcal{P}(1)$ holds. Hence

$$\mathcal{P}(p) \quad \text{holds for every} \quad p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

It remains to show that no other probability works. Since p is a probability, $p \in [0, 1]$. The complement of

$$[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}$$

inside $[0, 1]$ is

$$\left(\frac{1}{3}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right).$$

For every $p \in (1/3, 1/2)$, Proposition 8 gives an explicit choice of weights for which the desired inequality fails. For every $p \in (1/2, 1)$, Proposition 9 gives an explicit choice of weights for which the desired inequality fails.

Therefore $\mathcal{P}(p)$ holds exactly for

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

This proves the theorem. □

References

[1] W. Hoeffding, *The extrema of the expected value of a function of independent random variables*, Annals of Mathematical Statistics **26** (1955), 268–275.

[2] K. Jogdeo and S. M. Samuels, *Monotone convergence of binomial probabilities and a generalization of Ramanujan's equation*, Annals of Mathematical Statistics **39** (1968), 1191–1195.

[3] S. M. Samuels, *On the number of successes in independent trials*, Annals of Mathematical Statistics **36** (1965), 1272–1278.

[4] A. Fernandez, *Convex Combinations of Low Probability Bernoulli Variables*, Math.StackExchange discussion, `https://math.stackexchange.com/questions/265163/convex-combinations-of-low-probability-bernoulli-variables`.

Review of Submission 3B

First Proof Second Batch
Reviewer 2
June 4–8, 2026

1 Review

The submission claims that the inequality holds for $p \in [0, 1/3] \cup \{1/2, 1\}$. The submitted proof crucially relies on two results. First, it reduces the inequality from arbitrary convex combinations to the uniform convex combination $w_1 = \ldots = w_m = \frac{1}{m}$ by way of a comparison theorem. This reduces the inequality to the well-studied problem of bounding the probability that a binomial random variable exceeds its mean. Second, it claims that this inequality is true and follows from results in the literature.

Correctness. The submission fails to provide enough justification for both crucial steps. For the comparison theorem (Lemma 3), the submission cites a paper of Hoeffding. But it is unclear how to translate Hoeffding's results into Lemma 3. Moreover, as stated, Lemma 3 cannot hold and we provide a counter-example below. The claimed result might be true if one replaces i.i.d real-valued random variables with Bernoulli random variables, but it is still not obvious how to prove this.

For the second fact, the submission cites a paper of Samuels and a paper of Jogdeo-Samuels to claim that

$$\Pr[\text{Binom}(n, p) \geq np] \geq p \quad \text{for } 0 \leq p \leq \frac{1}{3}.$$

However, it is unclear how this result comes from these two papers.

The submission correctly prove that the inequality does not universally hold for any of the values outside of $[0, 1/3] \cup \{1/2, 1\}$.

Novelty. The submission does not propose any new approaches or ideas not already known in the literature.

2 Recommendation

The submission should be **rejected**. The proposed proof strategy is very natural, but the submission makes no meaningful progress towards this strategy.

3 Submission with inline comments

Overview / Strategy

The problem is to determine for which probabilities p the displayed inequality holds uniformly over all finite convex weight vectors. The main positive argument has two reductions. First, Hoeffding's extremal theorem for convex combinations of i.i.d. random variables reduces the weighted problem to the special case of equal weights. In the Bernoulli case, equal weights give binomial averages. Second, a classical binomial mean-tail estimate of Jogdeo–Samuels type says that for $0 \le p \le 1/3$, every binomial average is at least its mean with probability at least p. These two facts are stated precisely in Section 1 and used in Proposition 5.

The remaining cases are elementary. At $p = 1/2$, the Bernoulli subset is uniformly distributed over all subsets, and pairing every subset with its complement gives probability at least $1/2$. At $p = 1$, the assertion is immediate. To show that there are no other probabilities, equal weights on three variables give counterexamples for every $p \in (1/3, 1/2)$, and equal weights on two variables give counterexamples for every $p \in (1/2, 1)$.

Definition 1. A Bernoulli(p) random variable is a random variable taking value 1 with probability p and value 0 with probability $1 - p$. For $p \in [0, 1]$, let $\mathcal{P}(p)$ denote the following assertion: for every integer $m \ge 1$, every choice of nonnegative real numbers

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and every family $v_1, \dots, v_m$ of i.i.d. Bernoulli(p) random variables, one has

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p.$$

Theorem 2 (Headline theorem: the original problem)**.** *The assertion $\mathcal{P}(p)$ holds exactly for*

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

Equivalently, the complete set of probabilities p for which

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p$$

holds for every finite convex weight vector $(w_1, \dots, w_m)$ and every i.i.d. Bernoulli(p) family $v_1, \dots, v_m$ is

$$\boxed{[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}}.$$

3.1 Classical one-dimensional inputs

The proof uses two standard one-dimensional results. They are quoted in the precise forms needed below. The first is Hoeffding's reduction theorem for convex combinations of i.i.d. random variables; the second is the binomial mean-tail estimate used to evaluate the equal-weight case.

Lemma 3 (Hoeffding's reduction for convex combinations)**.** *Let $X_1, X_2, \ldots$ be i.i.d. real-valued random variables. For $t \in \mathbb{R}$, define*

$$L(t) := \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq t\right].$$

Then for every integer $m \geq 1$ and every choice of weights

$$w_1, \ldots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

one has

$$\Pr\left[\sum_{i=1}^{m} w_i X_i \geq t\right] \geq L(t).$$

Proof. This is the one-dimensional half-line form of Hoeffding's extremal theorem for convex combinations of i.i.d. random variables; see Hoeffding [1]. To make explicit that the closed-tail convention used here causes no ambiguity, we record the standard passage from the open-tail formulation.

> **Reviewer 2**
>
> It is unclear how the paper of Hoeffding gives the open-tail bound. Moreover, as stated, this result is incorrect. For a counter-example, consider the distribution $\Pr(X = 0) = \Pr(X = 1) = 0.1$ and $\Pr(X = 2) = 0.8$. Then
>
> $$\Pr\left[0.1X_1 + 0.1X_2 + 0.8X_3 > 0.8\right] = 0.899,$$
> $$\inf_{N \geq 1} \Pr\left[\frac{X_1 + \ldots + X_N}{N} \geq t\right] = 0.9$$
>
> It might be true that the result holds for Bernoulli random variables, but this is not obvious and is equivalent to Csóka's conjecture.

Hoeffding's theorem gives, for every real u,

$$\Pr\left[\sum_{i=1}^{m} w_i X_i > u\right] \geq \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} > u\right].$$

Fix $t \in \mathbb{R}$. For every $\varepsilon > 0$, apply the preceding inequality with $u = t - \varepsilon$. For each N,

$$\left\{\frac{X_1 + \cdots + X_N}{N} \geq t\right\} \subseteq \left\{\frac{X_1 + \cdots + X_N}{N} > t - \varepsilon\right\},$$

and therefore

$$\inf_{N\geq 1}\Pr\left[\frac{X_1+\cdots+X_N}{N}>t-\varepsilon\right]\geq\inf_{N\geq 1}\Pr\left[\frac{X_1+\cdots+X_N}{N}\geq t\right]=L(t).$$

Thus

$$\Pr\left[\sum_{i=1}^{m}w_iX_i>t-\varepsilon\right]\geq L(t)\qquad\text{for every }\varepsilon>0.$$

As $\varepsilon\downarrow 0$, the events

$$\left\{\sum_{i=1}^{m}w_iX_i>t-\varepsilon\right\}$$

decrease to

$$\left\{\sum_{i=1}^{m}w_iX_i\geq t\right\}.$$

By continuity of probability from above,

$$\Pr\left[\sum_{i=1}^{m}w_iX_i\geq t\right]\geq L(t).$$

This is the desired closed-tail statement. □

Lemma 4 (Binomial mean-tail inequality)**.** *Let*

$$X_{N,p}\sim\mathrm{Bin}(N,p).$$

Then the following hold.

1. *If $0\leq p\leq\frac{1}{3}$, then for every $N\geq 1$,*

$$\Pr[X_{N,p}\geq Np]\geq p.$$

   Equivalently, since $X_{N,p}$ is integer-valued,

$$\Pr[X_{N,p}\geq\lceil Np\rceil]\geq p.$$

2. *If $p=\frac{1}{2}$, then for every $N\geq 1$,*

$$\Pr[X_{N,1/2}\geq N/2]\geq\frac{1}{2}.$$

Moreover, in the first assertion equality occurs for $N=1$ and $0<p\leq 1/3$.

Proof. For $p=0$, $X_{N,0}=0$ almost surely, so

$$\Pr[X_{N,0}\geq 0]=1\geq 0.$$

For $0<p\leq 1/3$, the first assertion is the classical binomial mean-tail estimate of Samuels and Jogdeo–Samuels; see [3, 2].

> **Reviewer 2**
> It is not clear how the cited papers imply $\Pr[X_{N,p} \geq Np] \geq p$. There are papers of Greenberg–Mohri, Pelekis–Ramon, Doerr, Rigolett-Tong which study the question of bounding this quantity. It is unclear if this inequality follows from their works. For example, from Rigolett-Tong, we get
>
> $$\Pr[X_{N,p} \geq Np] \geq \min\left\{\frac{1}{4}, p\right\}$$
>
> but this is not enough.

In the form needed here, that estimate states exactly that for every $N \geq 1$,

$$\Pr[X_{N,p} \geq \lceil Np \rceil] \geq p.$$

Because $X_{N,p}$ is integer-valued, the events

$$\{X_{N,p} \geq Np\} \quad \text{and} \quad \{X_{N,p} \geq \lceil Np \rceil\}$$

are identical. This proves the first assertion.

For $p = 1/2$, the binomial distribution is symmetric:

$$\Pr[X_{N,1/2} = j] = \Pr[X_{N,1/2} = N - j] \qquad (0 \leq j \leq N).$$

If $N = 2r + 1$ is odd, then the values $0, \ldots, r$ pair with the values $2r + 1, \ldots, r + 1$, and exactly one member of each pair is at least $N/2 = r + \frac{1}{2}$. Hence

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}.$$

If $N = 2r$ is even, then the values $0, \ldots, r - 1$ pair with $2r, \ldots, r + 1$, while r is the central value and is included in the event $\{X_{N,1/2} \geq r\}$. Therefore

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}\Big(1 - \Pr[X_{N,1/2} = r]\Big) + \Pr[X_{N,1/2} = r] = \frac{1}{2} + \frac{1}{2}\Pr[X_{N,1/2} = r] \geq \frac{1}{2}.$$

This proves the second assertion.

Finally, if $N = 1$ and $0 < p \leq 1/3$, then

$$\{X_{1,p} \geq p\} = \{X_{1,p} = 1\},$$

so

$$\Pr[X_{1,p} \geq p] = p.$$

Thus equality is possible in the first assertion. $\square$

3.2 The positive cases

Proposition 5 (The interval $0 \le p \le 1/3$)**.** *The assertion $\mathcal{P}(p)$ holds for every*

$$p \in [0, \tfrac{1}{3}].$$

Proof. Fix $p \in [0, 1/3]$. Let $m \ge 1$, let

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables.

Extend $v_1, \dots, v_m$, if necessary, to an infinite i.i.d. Bernoulli(p) sequence $X_1, X_2, \dots$ with $X_i = v_i$ for $1 \le i \le m$. Applying Lemma 3 with $t = p$ gives

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] = \Pr\left[\sum_{i=1}^{m} w_i X_i \ge p\right] \ge \inf_{N \ge 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \ge p\right].$$

For each N, the sum $X_1 + \cdots + X_N$ has binomial distribution $\mathrm{Bin}(N, p)$. Hence, writing $X_{N,p} \sim \mathrm{Bin}(N, p)$,

$$\Pr\left[\frac{X_1 + \cdots + X_N}{N} \ge p\right] = \Pr[X_{N,p} \ge Np].$$

Therefore

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge \inf_{N \ge 1} \Pr[X_{N,p} \ge Np].$$

By Lemma 4, since $0 \le p \le 1/3$,

$$\Pr[X_{N,p} \ge Np] \ge p \qquad \text{for every } N \ge 1.$$

Taking the infimum over N gives

$$\inf_{N \ge 1} \Pr[X_{N,p} \ge Np] \ge p.$$

Consequently,

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \ge p\right] \ge p.$$

Since m, the weights, and the Bernoulli(p) family were arbitrary, $\mathcal{P}(p)$ holds. $\square$

Proposition 6 (The point $p = 1/2$)**.** *The assertion $\mathcal{P}(1/2)$ holds.*

Proof. Let $m \ge 1$, let

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli$(1/2)$ random variables.

Let
$$A = \{i \in \{1, \dots, m\} : v_i = 1\}.$$
Because the v_i are independent and each is 1 with probability $1/2$, the random set A is uniformly distributed over all 2^m subsets of $\{1, \dots, m\}$. For a subset $B \subseteq \{1, \dots, m\}$, write
$$w(B) = \sum_{i \in B} w_i.$$
Then
$$\sum_{i=1}^{m} w_i v_i = w(A),$$
so the desired event is
$$\{w(A) \geq 1/2\}.$$
Pair every subset $B \subseteq \{1, \dots, m\}$ with its complement B^c. These pairs partition the 2^m subsets into 2^{m-1} unordered pairs. Since
$$w(B) + w(B^c) = \sum_{i=1}^{m} w_i = 1,$$
at least one of B and B^c has weight at least $1/2$. Thus at least one subset in each complementary pair belongs to the event $\{w(A) \geq 1/2\}$.

Every subset has probability 2^{-m}. Since at least 2^{m-1} subsets satisfy $w(B) \geq 1/2$, we obtain
$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq \frac{1}{2}\right] = \Pr[w(A) \geq 1/2] \geq 2^{m-1} \cdot 2^{-m} = \frac{1}{2}.$$
Therefore $\mathcal{P}(1/2)$ holds. $\square$

Proposition 7 (The point $p = 1$)**.** *The assertion $\mathcal{P}(1)$ holds.*

Proof. Let $m \geq 1$, let
$$w_1, \dots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$
and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(1) random variables. A Bernoulli(1) random variable equals 1 almost surely, so
$$v_i = 1 \qquad \text{almost surely for every } i.$$
Therefore
$$\sum_{i=1}^{m} w_i v_i = \sum_{i=1}^{m} w_i = 1$$
almost surely. Hence
$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq 1\right] = 1 \geq 1.$$
Thus $\mathcal{P}(1)$ holds. $\square$

3.3 Counterexamples outside the claimed set

Proposition 8 (Failure for $1/3 < p < 1/2$)**.** *For every*

$$p \in \left(\frac{1}{3}, \frac{1}{2}\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/3, 1/2)$. Take

$$m = 3, \qquad w_1 = w_2 = w_3 = \frac{1}{3},$$

and let v_1, v_2, v_3 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2 + v_3.$$

Then $N \sim \mathrm{Bin}(3, p)$ and

$$\sum_{i=1}^{3} w_i v_i = \frac{N}{3}.$$

Because $p > 1/3$, one success is not enough:

$$\frac{1}{3} < p.$$

Because $p < 1/2 < 2/3$, two successes are enough:

$$\frac{2}{3} \geq p.$$

Thus

$$\left\{\sum_{i=1}^{3} w_i v_i \geq p\right\} = \{N \geq 2\}.$$

Therefore

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] = \Pr[N \geq 2].$$

Since $N \sim \mathrm{Bin}(3, p)$,

$$\Pr[N \geq 2] = \binom{3}{2} p^2(1-p) + p^3 = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p - p^2(3-2p) = p(1 - 3p + 2p^2) = p(1-p)(1-2p).$$

For $p \in (1/3, 1/2)$, the factors p, $1-p$, and $1-2p$ are all positive. Hence

$$p - p^2(3-2p) > 0,$$

or equivalently,

$$p^2(3-2p) < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

Proposition 9 (Failure for $1/2 < p < 1$)**.** *For every*

$$p \in \left(\frac{1}{2}, 1\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/2, 1)$. Take

$$m = 2, \qquad w_1 = w_2 = \frac{1}{2},$$

and let v_1, v_2 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2.$$

Then

$$\sum_{i=1}^{2} w_i v_i = \frac{N}{2}.$$

Because $p > 1/2$, one success is not enough:

$$\frac{1}{2} < p.$$

Since N can only take the values $0, 1, 2$, the event

$$\left\{\frac{N}{2} \geq p\right\}$$

occurs only when $N = 2$, that is, when both v_1 and v_2 equal 1. Hence

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] = \Pr[v_1 = 1,\ v_2 = 1].$$

By independence,

$$\Pr[v_1 = 1,\ v_2 = 1] = p^2.$$

Since $0 < p < 1$,

$$p^2 < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

3.4 Proof of the classification

Proof of Theorem 2. Proposition 5 proves that $\mathcal{P}(p)$ holds for every

$$p \in [0, \tfrac{1}{3}].$$

Proposition 6 proves that $\mathcal{P}(1/2)$ holds, and Proposition 7 proves that $\mathcal{P}(1)$ holds. Hence

$$\mathcal{P}(p) \quad \text{holds for every} \quad p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

It remains to show that no other probability works. Since p is a probability, $p \in [0, 1]$. The complement of

$$[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}$$

inside $[0, 1]$ is

$$\left(\frac{1}{3}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right).$$

For every $p \in (1/3, 1/2)$, Proposition 8 gives an explicit choice of weights for which the desired inequality fails. For every $p \in (1/2, 1)$, Proposition 9 gives an explicit choice of weights for which the desired inequality fails.

Therefore $\mathcal{P}(p)$ holds exactly for

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

This proves the theorem. □

References

[1] W. Hoeffding, *The extrema of the expected value of a function of independent random variables*, Annals of Mathematical Statistics **26** (1955), 268–275.

[2] K. Jogdeo and S. M. Samuels, *Monotone convergence of binomial probabilities and a generalization of Ramanujan's equation*, Annals of Mathematical Statistics **39** (1968), 1191–1195.

[3] S. M. Samuels, *On the number of successes in independent trials*, Annals of Mathematical Statistics **36** (1965), 1272–1278.

[4] A. Fernandez, *Convex Combinations of Low Probability Bernoulli Variables*, Math.StackExchange discussion, `https://math.stackexchange.com/questions/265163/convex-combinations-of-low-probability-bernoulli-variables`.

Review of Submission 3B

First Proof Second Batch
Reviewer 3
June 4–8, 2026

1 Review

Correctness. My recommendation is that the submission **should be rejected**. The proof attempt relies on two key inequalities, Lemma 3 and Lemma 4.

The proof of Lemma 3 is based on a false claim. This constitutes a fundamental flaw in the proof. Lemma 4, meanwhile, is not stated in the cited sources. The submission claims that it is a classical result following from [3] and [2], but neither paper contains the lemma in the form stated here. While it is plausible that a result of this type could be derived from those works, such a derivation is not immediate, and no justification is provided.

The remainder of the argument, which reduces the main claim to these two lemmas, appears to be correct.

Novelty. I would say that the novelty of this solution is low. Even assuming that Lemmas were correct, the main argument consists essentially of an immediate reduction to these auxiliary results.

2 Recommendation

3 Submission with inline comments

Overview / Strategy

The problem is to determine for which probabilities p the displayed inequality holds uniformly over all finite convex weight vectors. The main positive argument has two reductions. First, Hoeffding's extremal theorem for convex combinations of i.i.d. random variables reduces the weighted problem to the special case of equal weights. In the Bernoulli case, equal weights give binomial averages. Second, a classical binomial mean-tail estimate of Jogdeo–Samuels type says that for $0 \leq p \leq 1/3$, every binomial average is at least its mean with probability at least p. These two facts are stated precisely in Section 1 and used in Proposition 5.

The remaining cases are elementary. At $p = 1/2$, the Bernoulli subset is uniformly distributed over all subsets, and pairing every subset with its complement gives probability at least $1/2$. At $p = 1$, the assertion is immediate. To show that there are no other probabilities, equal weights on three variables give counterexamples for every $p \in (1/3, 1/2)$, and equal weights on two variables give counterexamples for every $p \in (1/2, 1)$.

Definition 1. A Bernoulli(p) random variable is a random variable taking value 1 with probability p and value 0 with probability $1 - p$. For $p \in [0, 1]$, let $\mathcal{P}(p)$ denote the following assertion: for every integer $m \geq 1$, every choice of nonnegative real numbers

$$w_1, \ldots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and every family $v_1, \ldots, v_m$ of i.i.d. Bernoulli(p) random variables, one has

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq p\right] \geq p.$$

Theorem 2 (Headline theorem: the original problem)**.** *The assertion $\mathcal{P}(p)$ holds exactly for*

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

Equivalently, the complete set of probabilities p for which

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq p\right] \geq p$$

holds for every finite convex weight vector $(w_1, \ldots, w_m)$ and every i.i.d. Bernoulli(p) family $v_1, \ldots, v_m$ is

$$\boxed{[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}}.$$

3.1 Classical one-dimensional inputs

The proof uses two standard one-dimensional results. They are quoted in the precise forms needed below. The first is Hoeffding's reduction theorem for convex combinations of i.i.d. random variables; the second is the binomial mean-tail estimate used to evaluate the equal-weight case.

Lemma 3 (Hoeffding's reduction for convex combinations)**.** *Let $X_1, X_2, \ldots$ be i.i.d. real-valued random variables. For $t \in \mathbb{R}$, define*

$$L(t) := \inf_{N \geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq t\right].$$

Then for every integer $m \geq 1$ and every choice of weights

$$w_1, \ldots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

one has

$$\Pr\left[\sum_{i=1}^{m} w_i X_i \geq t\right] \geq L(t).$$

Proof. This is the one-dimensional half-line form of Hoeffding's extremal theorem for convex combinations of i.i.d. random variables; see Hoeffding [1]. To make explicit that the closed-tail convention used here causes no ambiguity, we record the standard passage from the open-tail formulation.

Hoeffding's theorem gives, for every real u,

$$\Pr\left[\sum_{i=1}^{m} w_i X_i > u\right] \geq \inf_{N\geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} > u\right].$$

> **Reviewer 3**
>
> No such statement appears in Hoeffding [1]. In fact, the claim is wrong as stated. It has a counterexample already for $m = 2$.

Fix $t \in \mathbb{R}$. For every $\varepsilon > 0$, apply the preceding inequality with $u = t - \varepsilon$. For each N,

$$\left\{\frac{X_1 + \cdots + X_N}{N} \geq t\right\} \subseteq \left\{\frac{X_1 + \cdots + X_N}{N} > t - \varepsilon\right\},$$

and therefore

$$\inf_{N\geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} > t - \varepsilon\right] \geq \inf_{N\geq 1} \Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq t\right] = L(t).$$

Thus

$$\Pr\left[\sum_{i=1}^{m} w_i X_i > t - \varepsilon\right] \geq L(t) \qquad \text{for every } \varepsilon > 0.$$

As $\varepsilon \downarrow 0$, the events

$$\left\{\sum_{i=1}^{m} w_i X_i > t - \varepsilon\right\}$$

decrease to

$$\left\{\sum_{i=1}^{m} w_i X_i \geq t\right\}.$$

By continuity of probability from above,

$$\Pr\left[\sum_{i=1}^{m} w_i X_i \geq t\right] \geq L(t).$$

This is the desired closed-tail statement. □

Lemma 4 (Binomial mean-tail inequality)**.** *Let*

$$X_{N,p} \sim \mathrm{Bin}(N, p).$$

Then the following hold.

1. *If* $0 \le p \le \frac{1}{3}$*, then for every* $N \ge 1$*,*

$$\Pr[X_{N,p} \ge Np] \ge p.$$

Equivalently, since $X_{N,p}$ *is integer-valued,*

$$\Pr[X_{N,p} \ge \lceil Np \rceil] \ge p.$$

2. *If* $p = \frac{1}{2}$*, then for every* $N \ge 1$*,*

$$\Pr[X_{N,1/2} \ge N/2] \ge \frac{1}{2}.$$

Moreover, in the first assertion equality occurs for $N = 1$ *and* $0 < p \le 1/3$*.*

Proof. For $p = 0$, $X_{N,0} = 0$ almost surely, so

$$\Pr[X_{N,0} \ge 0] = 1 \ge 0.$$

For $0 < p \le 1/3$, the first assertion is the classical binomial mean-tail estimate of Samuels and Jogdeo–Samuels; see [3, 2]. In the form needed here, that estimate states exactly that for every $N \ge 1$,

$$\Pr[X_{N,p} \ge \lceil Np \rceil] \ge p.$$

Because $X_{N,p}$ is integer-valued, the events

$$\{X_{N,p} \ge Np\} \quad \text{and} \quad \{X_{N,p} \ge \lceil Np \rceil\}$$

are identical. This proves the first assertion.

> **Reviewer 3**
>
> Neither [3] nor [2] contains this statement in the form presented here. While it is plausible that a bound of this type could be derived from the results in those papers, it is not immediately clear how such a derivation would proceed.

For $p = 1/2$, the binomial distribution is symmetric:

$$\Pr[X_{N,1/2} = j] = \Pr[X_{N,1/2} = N - j] \qquad (0 \le j \le N).$$

If $N = 2r + 1$ is odd, then the values $0, \dots, r$ pair with the values $2r + 1, \dots, r + 1$, and exactly one member of each pair is at least $N/2 = r + \frac{1}{2}$. Hence

$$\Pr[X_{N,1/2} \ge N/2] = \frac{1}{2}.$$

If $N = 2r$ is even, then the values $0, \dots, r - 1$ pair with $2r, \dots, r + 1$, while r is the central value and is included in the event $\{X_{N,1/2} \ge r\}$. Therefore

$$\Pr[X_{N,1/2} \ge N/2] = \frac{1}{2}\Big(1 - \Pr[X_{N,1/2} = r]\Big) + \Pr[X_{N,1/2} = r] = \frac{1}{2} + \frac{1}{2}\Pr[X_{N,1/2} = r] \ge \frac{1}{2}.$$

This proves the second assertion.

Finally, if $N = 1$ and $0 < p \le 1/3$, then

$$\{X_{1,p} \ge p\} = \{X_{1,p} = 1\},$$

so

$$\Pr[X_{1,p} \ge p] = p.$$

Thus equality is possible in the first assertion. $\square$

3.2 The positive cases

Proposition 5 (The interval $0 \le p \le 1/3$)**.** *The assertion $\mathcal{P}(p)$ holds for every*

$$p \in [0, \tfrac{1}{3}].$$

Proof. Fix $p \in [0, 1/3]$. Let $m \ge 1$, let

$$w_1, \dots, w_m \ge 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables.

Extend $v_1, \dots, v_m$, if necessary, to an infinite i.i.d. Bernoulli(p) sequence $X_1, X_2, \dots$ with $X_i = v_i$ for $1 \le i \le m$. Applying Lemma 3 with $t = p$ gives

$$\Pr\Bigg[\sum_{i=1}^{m} w_i v_i \ge p\Bigg] = \Pr\Bigg[\sum_{i=1}^{m} w_i X_i \ge p\Bigg] \ge \inf_{N \ge 1} \Pr\Big[\frac{X_1 + \cdots + X_N}{N} \ge p\Big].$$

For each N, the sum $X_1 + \cdots + X_N$ has binomial distribution $\mathrm{Bin}(N, p)$. Hence, writing $X_{N,p} \sim \mathrm{Bin}(N, p)$,

$$\Pr\Big[\frac{X_1 + \cdots + X_N}{N} \ge p\Big] = \Pr[X_{N,p} \ge Np].$$

Therefore

$$\Pr\Bigg[\sum_{i=1}^{m} w_i v_i \ge p\Bigg] \ge \inf_{N \ge 1} \Pr[X_{N,p} \ge Np].$$

By Lemma 4, since $0 \le p \le 1/3$,

$$\Pr[X_{N,p} \ge Np] \ge p \qquad \text{for every } N \ge 1.$$

Taking the infimum over N gives

$$\inf_{N \ge 1} \Pr[X_{N,p} \ge Np] \ge p.$$

Consequently,

$$\Pr\Bigg[\sum_{i=1}^{m} w_i v_i \ge p\Bigg] \ge p.$$

Since m, the weights, and the Bernoulli(p) family were arbitrary, $\mathcal{P}(p)$ holds. $\square$

Proposition 6 (The point $p = 1/2$)**.** *The assertion $\mathcal{P}(1/2)$ holds.*

Proof. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli$(1/2)$ random variables.

Let

$$A = \{i \in \{1, \dots, m\} : v_i = 1\}.$$

Because the v_i are independent and each is 1 with probability $1/2$, the random set A is uniformly distributed over all 2^m subsets of $\{1, \dots, m\}$. For a subset $B \subseteq \{1, \dots, m\}$, write

$$w(B) = \sum_{i \in B} w_i.$$

Then

$$\sum_{i=1}^{m} w_i v_i = w(A),$$

so the desired event is

$$\{w(A) \geq 1/2\}.$$

Pair every subset $B \subseteq \{1, \dots, m\}$ with its complement B^c. These pairs partition the 2^m subsets into 2^{m-1} unordered pairs. Since

$$w(B) + w(B^c) = \sum_{i=1}^{m} w_i = 1,$$

at least one of B and B^c has weight at least $1/2$. Thus at least one subset in each complementary pair belongs to the event $\{w(A) \geq 1/2\}$.

Every subset has probability 2^{-m}. Since at least 2^{m-1} subsets satisfy $w(B) \geq 1/2$, we obtain

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq \frac{1}{2}\right] = \Pr[w(A) \geq 1/2] \geq 2^{m-1} \cdot 2^{-m} = \frac{1}{2}.$$

Therefore $\mathcal{P}(1/2)$ holds. $\square$

Proposition 7 (The point $p = 1$)**.** *The assertion $\mathcal{P}(1)$ holds.*

Proof. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \qquad \sum_{i=1}^{m} w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(1) random variables. A Bernoulli(1) random variable equals 1 almost surely, so

$$v_i = 1 \qquad \text{almost surely for every } i.$$

Therefore

$$\sum_{i=1}^{m} w_i v_i = \sum_{i=1}^{m} w_i = 1$$

almost surely. Hence

$$\Pr\left[\sum_{i=1}^{m} w_i v_i \geq 1\right] = 1 \geq 1.$$

Thus $\mathcal{P}(1)$ holds. $\square$

3.3 Counterexamples outside the claimed set

Proposition 8 (Failure for $1/3 < p < 1/2$)**.** *For every*

$$p \in \left(\frac{1}{3}, \frac{1}{2}\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/3, 1/2)$. Take

$$m = 3, \qquad w_1 = w_2 = w_3 = \frac{1}{3},$$

and let v_1, v_2, v_3 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2 + v_3.$$

Then $N \sim \mathrm{Bin}(3, p)$ and

$$\sum_{i=1}^{3} w_i v_i = \frac{N}{3}.$$

Because $p > 1/3$, one success is not enough:

$$\frac{1}{3} < p.$$

Because $p < 1/2 < 2/3$, two successes are enough:

$$\frac{2}{3} \geq p.$$

Thus

$$\left\{\sum_{i=1}^{3} w_i v_i \geq p\right\} = \{N \geq 2\}.$$

Therefore

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] = \Pr[N \geq 2].$$

Since $N \sim \mathrm{Bin}(3, p)$,

$$\Pr[N \geq 2] = \binom{3}{2} p^2(1-p) + p^3 = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p - p^2(3-2p) = p(1 - 3p + 2p^2) = p(1-p)(1-2p).$$

For $p \in (1/3, 1/2)$, the factors p, $1-p$, and $1-2p$ are all positive. Hence

$$p - p^2(3-2p) > 0,$$

or equivalently,

$$p^2(3-2p) < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{3} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

Proposition 9 (Failure for $1/2 < p < 1$)**.** *For every*

$$p \in \left(\frac{1}{2}, 1\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/2, 1)$. Take

$$m = 2, \qquad w_1 = w_2 = \frac{1}{2},$$

and let v_1, v_2 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2.$$

Then

$$\sum_{i=1}^{2} w_i v_i = \frac{N}{2}.$$

Because $p > 1/2$, one success is not enough:

$$\frac{1}{2} < p.$$

Since N can only take the values $0, 1, 2$, the event

$$\left\{\frac{N}{2} \geq p\right\}$$

occurs only when $N = 2$, that is, when both v_1 and v_2 equal 1. Hence

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] = \Pr[v_1 = 1,\ v_2 = 1].$$

By independence,

$$\Pr[v_1 = 1,\ v_2 = 1] = p^2.$$

Since $0 < p < 1$,

$$p^2 < p.$$

Thus this choice of weights gives

$$\Pr\left[\sum_{i=1}^{2} w_i v_i \geq p\right] < p.$$

Therefore $\mathcal{P}(p)$ fails. $\square$

3.4 Proof of the classification

Proof of Theorem 2. Proposition 5 proves that $\mathcal{P}(p)$ holds for every

$$p \in [0, \tfrac{1}{3}].$$

Proposition 6 proves that $\mathcal{P}(1/2)$ holds, and Proposition 7 proves that $\mathcal{P}(1)$ holds. Hence

$$\mathcal{P}(p) \quad \text{holds for every} \quad p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

It remains to show that no other probability works. Since p is a probability, $p \in [0, 1]$. The complement of

$$[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}$$

inside $[0, 1]$ is

$$\left(\frac{1}{3}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right).$$

For every $p \in (1/3, 1/2)$, Proposition 8 gives an explicit choice of weights for which the desired inequality fails. For every $p \in (1/2, 1)$, Proposition 9 gives an explicit choice of weights for which the desired inequality fails.

Therefore $\mathcal{P}(p)$ holds exactly for

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

This proves the theorem. $\square$

References

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